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REVIEWERS: ARMY - MO
AIR FORCE - 85
- 69
USERS: NAVY - YD
- MC

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Other Cust.
NAVY - YD
AIR FORCE - 69
PROCUREMENT SPECIFICATION SUPERSEDES: MIL-S-3785

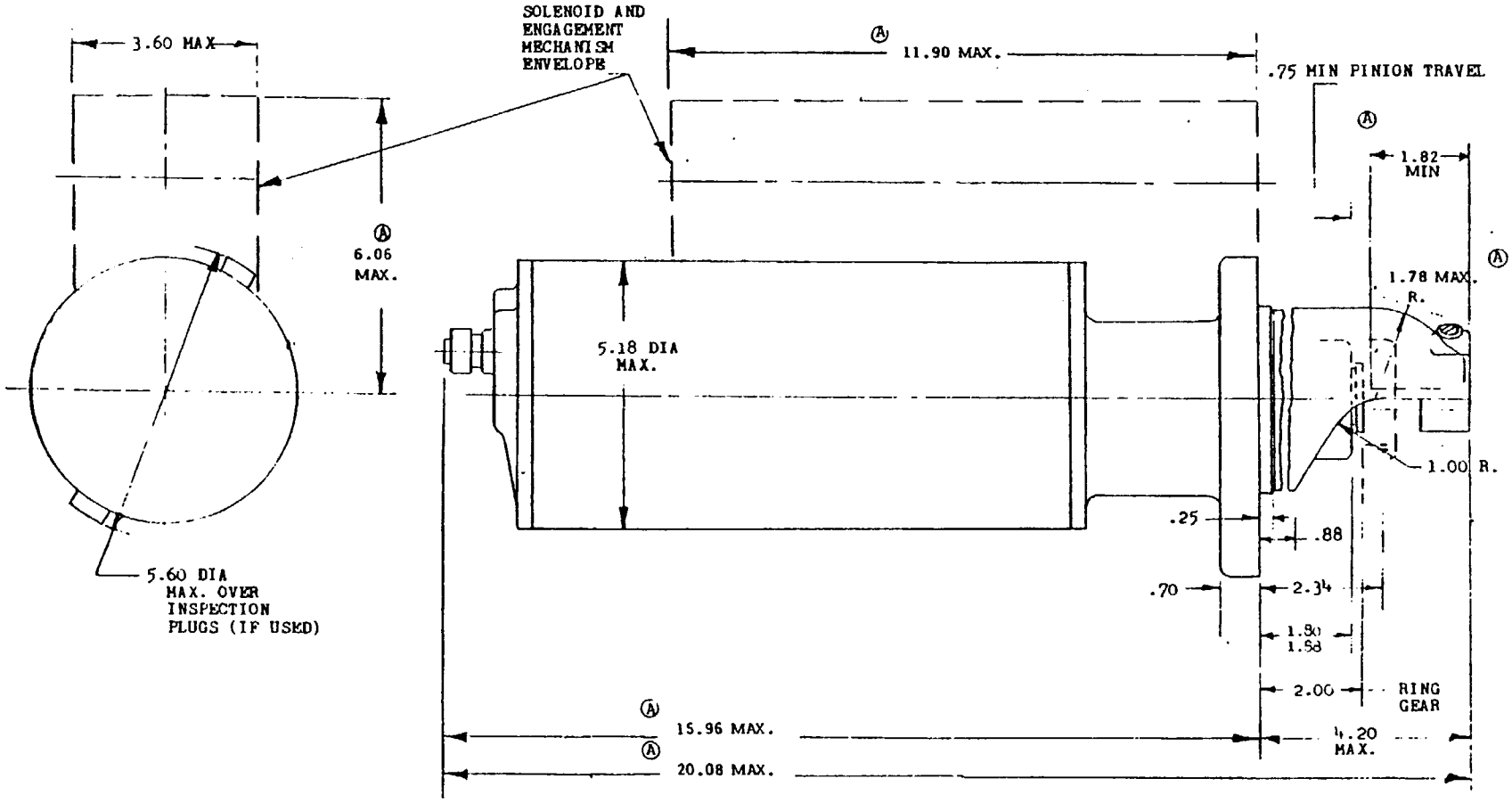
TITLE
Ⓐ STARTER, ENGINE, ELECTRICAL, 24-VOLT, D.C., 5-1/3-INCH FRAME DIAMETER SOLENOID-ACTUATED, HEAVY-DUTY

Ⓐ FOR CHANGES SEE SHEETS 1 THRU 5.

MILITARY STANDARD

MS 53009

SHEET 1 OF 5



NOTES:
1. PINION DATA
Ⓐ MS-1 AND -2: NO. OF TEETH (SEE TABLE); 8/10 PITCH; 1.625 P.D.; 1.825 O.D.; 20° PRESSURE ANGLE; .015 TO .025 P.L. CLEARANCE.
MS-3: NO. OF TEETH (SEE TABLE); 6/8 PITCH; 2.000 P.D.; 2.250 O.D.; 20° PRESSURE ANGLE; .015 TO .025 P.L. CLEARANCE.
2. MOTOR IS INSULATED TYPE MOTOR.
3. MOUNTING IS SIMILAR TO #3 SAE FLANGE MOUNTING.
Ⓐ 4. TERMINAL THREAD SIZE AND LENGTH: BATTERY - 1/2 - 13 UNC - 2A, MINIMUM LENGTH - 1.50
MOTOR-GROUND - 1/2 - 13 UNC - 2A, MINIMUM LENGTH - 1.00
SWITCH - #10 - 32 UNF 2A
5. DESIGN AND ARRANGEMENT OF SOLENOID AND ENGAGEMENT MECHANISM ASSEMBLY ARE OPTIONAL, PROVIDED SHIFT LEVER AND LINKAGE ARE INCLOSED AND ALSO PROVIDED TERMINAL THREAD SIZES CONFORM TO AND LIMITING OVERALL DIMENSIONS DO NOT EXCEED THOSE SPECIFIED HEREIN.
Ⓐ 6. ROTATION VIEWING DRIVE END - CLOCKWISE.
Ⓐ 7. TOLERANCE: DIMENSIONS; ± .030 UNLESS OTHERWISE SPECIFIED. ANGULAR; ± 1°.

(NOTES CONTINUED ON SHEET 2)

MS PART NO.	TEETH	PERFORMANCE CURVE
MS 53009-1	12T - 13T BLANK	SEE SHEET 4
MS 53009-2	12T - 13T BLANK	SEE SHEET 5
MS 53009-3	11T - 12T BLANK	SEE SHEET 5

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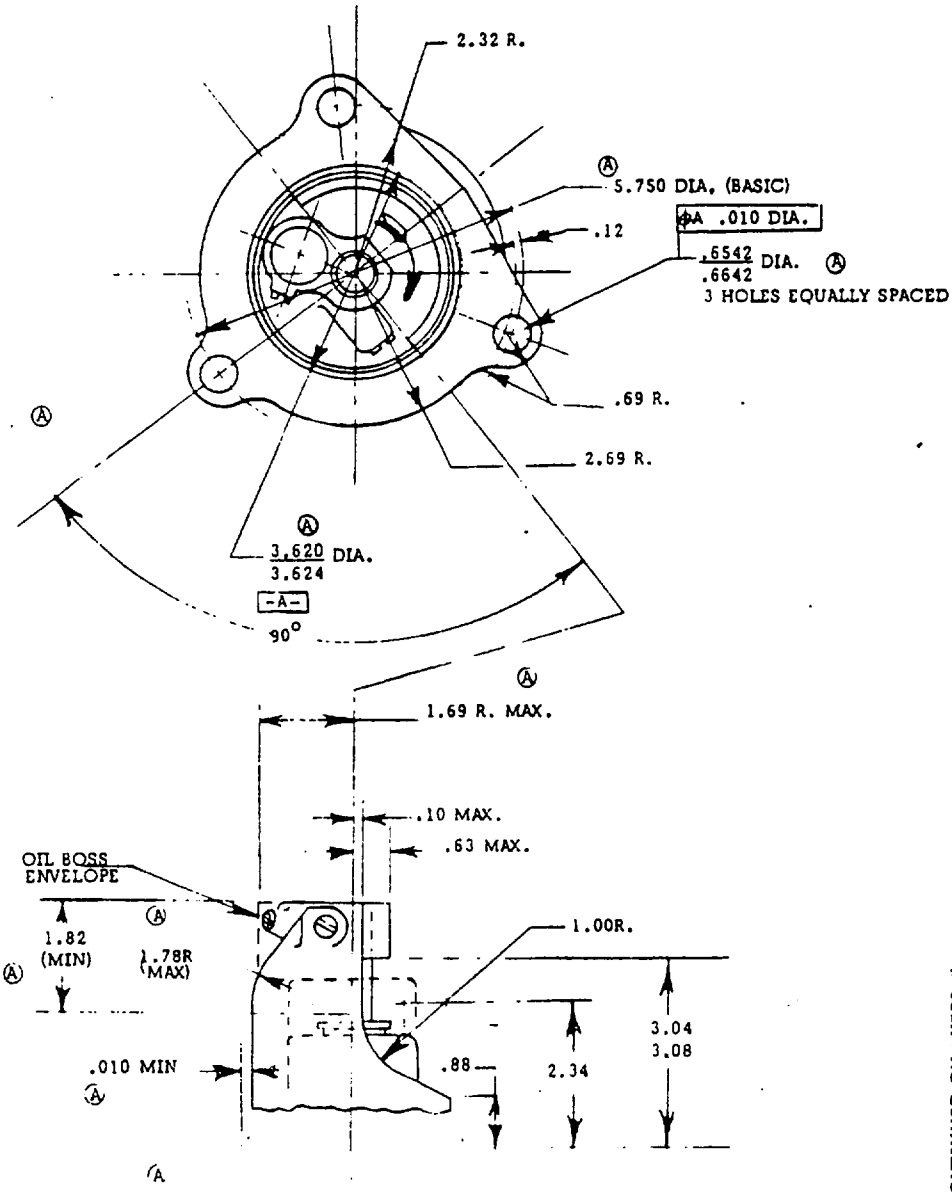
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(CONTINUATION OF NOTES FROM SHEET 1)

8. THIS MOTOR IS IDENTIFIED AS TYPE II (RING GEAR ENGAGEMENT), GRADE A (WATERPROOF), CLASS 1 (POSITIVE DRIVE) OF REFERENCED SPECIFICATION.
9. MOTOR SHALL BE EQUIPPED WITH OIL SEAL FOR WET CLUTCH OPERATION.
10. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.
11. MODIFICATIONS IN OUTSIDE APPEARANCE AND CONSTRUCTION ARE ACCEPTABLE IF UNIT INTERCHANGEABILITY IS NOT AFFECTED.
Ⓐ 12. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.
Ⓐ 13. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

(NOTES CONTINUED ON SHEET 4)

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P.A. ARMY - MO(ERDL) Other Cust. NAVY - YD AIR FORCE - 69	TITLE Ⓐ STARTER, ENGINE, ELECTRICAL, 24-VOLT, D.C., 5-1/8-INCH FRAME DIAMETER, SOLENOID-ACTUATED, HEAVY-DUTY	MILITARY STANDARD MS 53009
PROCUREMENT SPECIFICATION SUPERSEDES: MIL-S-3785		SHEET 2

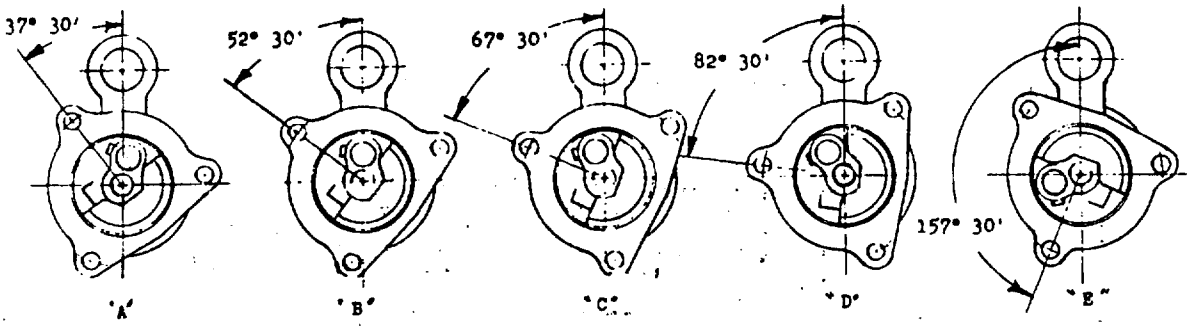
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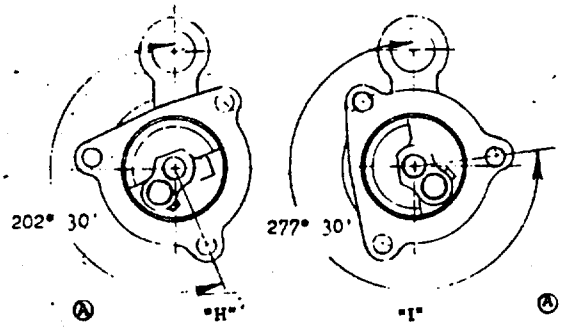
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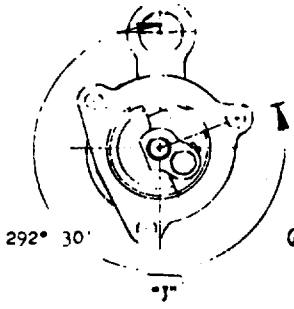
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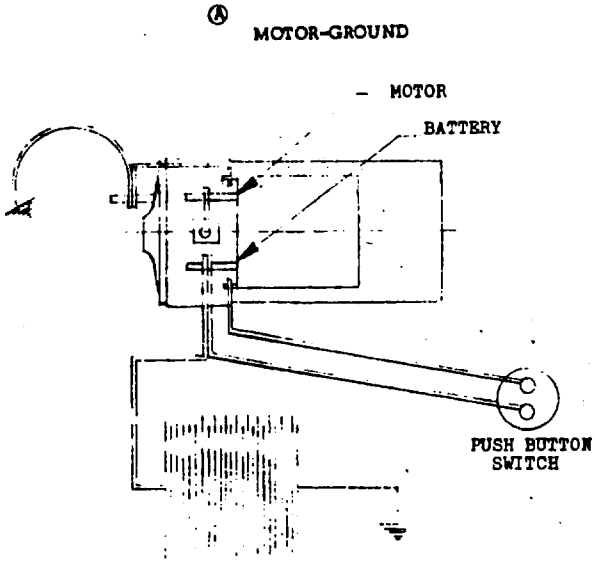




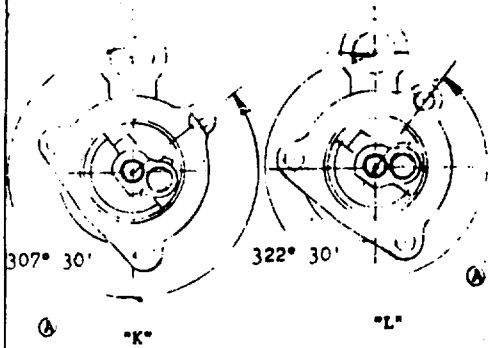
MOUNTING FLANGE AND PINION HOUSING SHALL BE ADJUSTABLE TO ALL POSITIONS SHOWN HEREON. MOUNTING POSITIONS VIEWING DRIVE END OF MOTOR.



FIGURES "F" AND "G" HAVE BEEN DELETED



TYPICAL WIRING DIAGRAM



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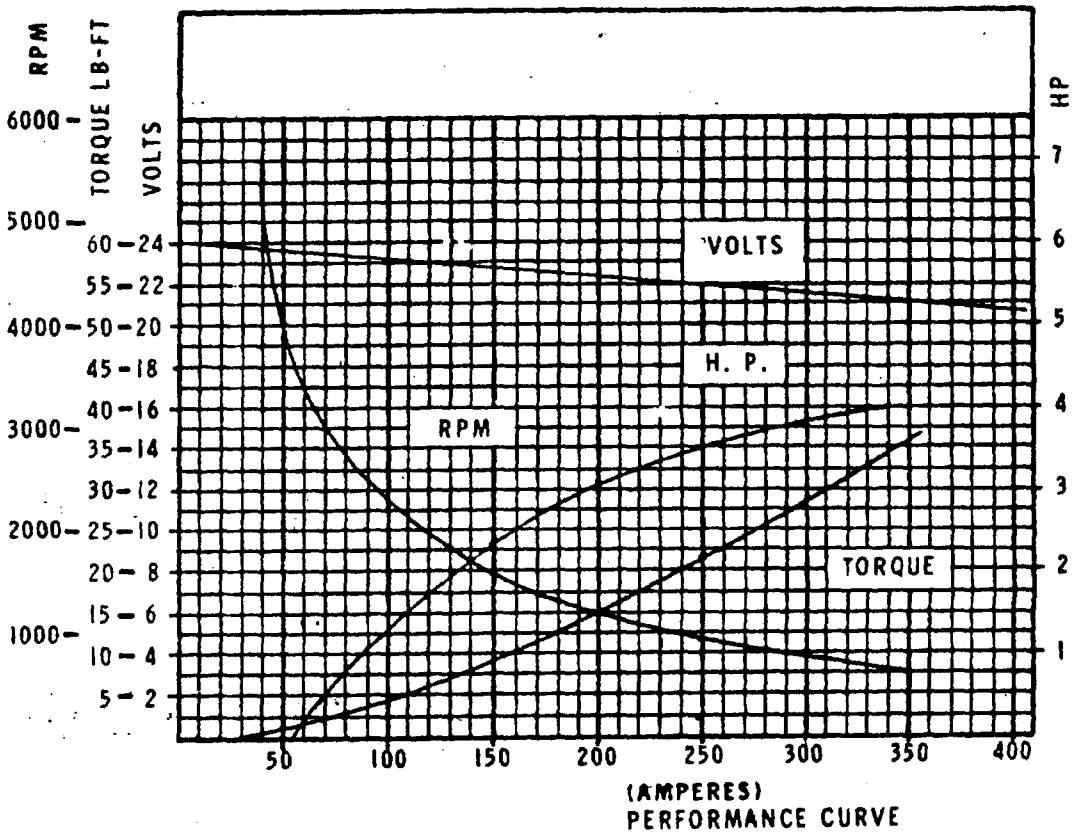
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P.A. ARMY - MO(ERDL) Other Cust. NAVY - YD AIR FORCE - 69	**MILITARY STANDARD** Ⓐ STARTER, ENGINE, ELECTRICAL, 24-VOLT, D.C., 5-1/8-INCH FRAME DIAMETER, SOLENOID-ACTUATED, HEAVY-DUTY	**MS 53009**
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 3

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PERFORMANCE CURVE
MS 53009-1 STARTER

(CONTINUATION OF NOTES FROM SHEET 2)

14. TORQUE AND R.P.M. MUST NOT BE LESS THAN 10% BELOW THAT SHOWN BY CURVES WITH VOLTAGE APPLIED AS SPECIFIED. OUTPUT CURVE IS NOT RELATED TO TORQUE AND R.P.M. CURVES SHOWN, BUT IS THE MINIMUM FOR ANY PARTICULAR MOTOR.
15. THE VOLTAGE CURVE SHOWN IS THE SAE HEAVY DUTY 24-VOLT BATTERY CURVE.
16. LOCATION OF NAMEPLATE SHALL BE OPTIONAL. NAMEPLATE SHALL BE FASTENED TO STARTER FRAME WITH RIVETS OR SCREWS AND LOCK WASHERS.
17. STALLED TORQUE DATA: VOLTS - 6.6 (APPROX)
CURRENT (AMPS) (MAX) - 200
TORQUE (LB-FT) (MIN) - 12
18. JUMPER BETWEEN SOLENOID-GROUND TERMINAL AND MOTOR-GROUND TERMINAL SHALL BE FURNISHED WITH THIS STARTER.
Ⓐ 19. SOLENOID CURRENT CONSUMPTION (PULL-IN PLUS HOLD-IN) SHALL NOT EXCEED 61 AMPERES.

Ⓐ REDRAWN

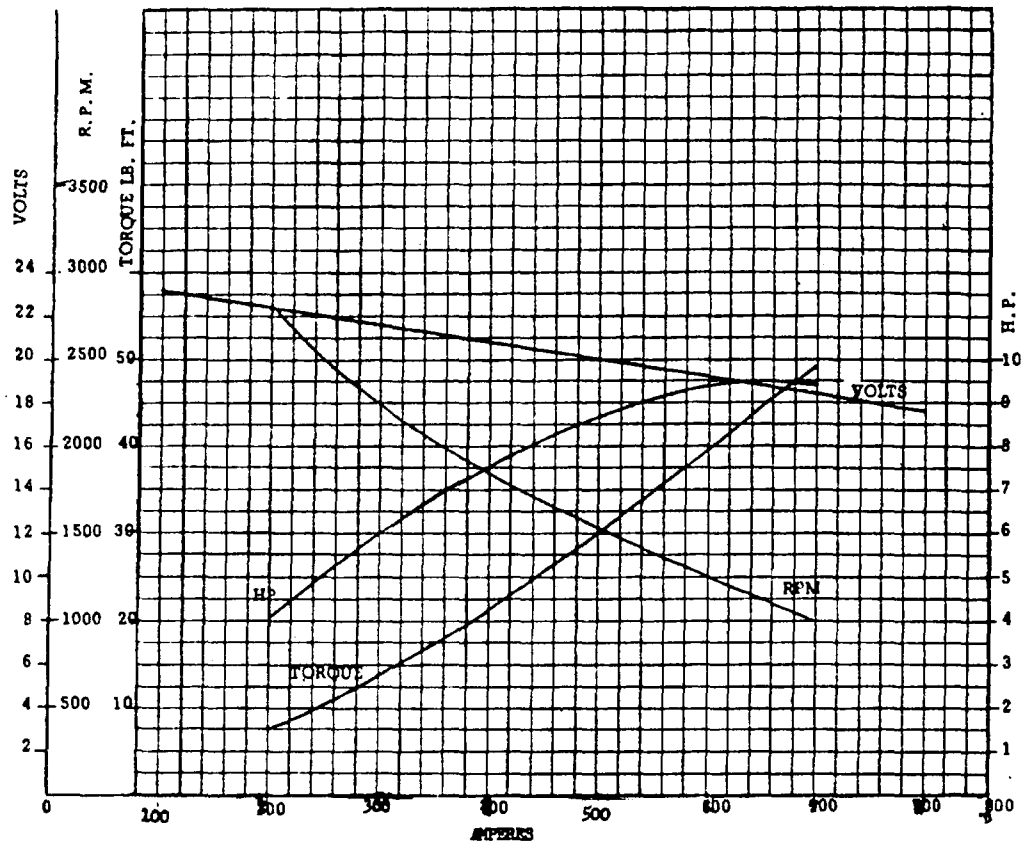
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P.A. ARMY - MO(ERDL) Other Cust. NAVY - YD AIR FORCE - 69	TITLE STARTER, ENGINE, ELECTRICAL, 24-VOLT, D.C., 5-1/8-INCH FRAME DIAMETER, SOLENOID-ACTUATED, HEAVY-DUTY	MILITARY STANDARD MS 53009
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 4

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PERFORMANCE CURVE

MS 53009-2 AND -3 STARTER

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Ⓐ SHEET ADDED

P.A. ARMY - MO(ERDL) Other Cust. NAVY - YD AIR FORCE - 69	TITLE STARTER, ENGINE, ELECTRICAL, 24-VOLT, D.C., 5-1/8-INCH FRAME DIAMETER SOLENOID-ACTUATED, HEAVY-DUTY	MILITARY STANDARD MS 53009
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 5 OF

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